

How far has the diesel pool moved from the certification fuel? A state-level specification-based atlas for the United States, 2011–2024

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Abstract

Compression-ignition engines sold in the United States are certified on a test fuel fixed by 40 CFR 1065.703. The fuel they subsequently burn is whatever the market supplies, and the market has changed: fatty acid methyl ester biodiesel and hydrotreated renewable diesel now account for a growing share of transportation distillate, and that share is distributed across states extremely unevenly.

No federal programme publishes measurements of in-service diesel properties. This paper therefore

asks a question that public data can answer: **given what each state consumed, what do the governing fuel specifications imply about how far its diesel pool sits from the certification fuel?**

We combine Energy Information Administration state-level consumption volumes for 2011–2024 (728 state-years, 51 jurisdictions) with published specification envelopes for the three pool components, and propagate the full envelopes rather than only their midpoints, so every estimate carries an interval that reflects what the specifications actually permit.

The headline results are asymmetric, and the asymmetry is the contribution. For states blending hydrotreated renewable diesel, the **direction** of divergence is established for any admissible property values: HVO’s density envelope (765–800 kg/m³) lies entirely below the certification fuel’s (838.9–864.6 kg/m³) and its cetane envelope (70–80) entirely above (40–50). California’s 2024 pool is **54.6% non-petroleum** — 51% renewable diesel — and sits **34 kg/m³ lighter** than the certification fuel, with a sensitivity interval of [-50.8, -17.4] that excludes zero.

For states blending only FAME biodiesel, **the direction is not established**. FAME’s density and cetane envelopes both overlap the certification fuel’s, so the sign of the deviation depends on where inside its permitted range the actual fuel sits. Across the panel, the density deviation is sign-robust in only **2.7%** of state-years and the cetane deviation in **4.7%** — essentially the HVO-blending states alone. Midpoint estimates do show biodiesel states heavier, and that may be true; the specifications do not establish it and this paper does not claim it.

We also document a methodological trap in the source data. EIA changed its distillate accounting at data year 2021; the intuitive blend-share construction double-counts from that year forward while being correct before, inflating the national denominator by about 20,008 thousand barrels a year (1.726%) and manufacturing a trend break that looks like a change in the fuel supply. The pipeline detects the break from the data and aborts if it is not where it is expected.

Every state-year carries its sensitivity interval and two sign-robustness flags, so the distinction between what this dataset establishes and what it merely suggests is carried by the data rather than by a limitations section.

Keywords: diesel fuel; renewable diesel; biodiesel; engine certification; fuel specifications; EIA SEDS; uncertainty propagation; open government data

1 Introduction

1.1 The question and why public data can answer only part of it

A compression-ignition engine is certified once, on a test fuel fixed by 40 CFR 1065.703, and then operated for thousands of hours on fuel drawn from the market. A companion study in the same research program establishes that the in-market fuel requirement leaves most injection-relevant fuel properties unconstrained, and that the federal government publishes measurements of none of them.

That result creates an evidentiary problem rather than solving one. If nobody measures in-service fuel properties, how far the fuel has actually moved is not a question public data can answer directly.

It can, however, answer a narrower question. Consumption volumes by component are published; specification envelopes for each component are published. Combining them yields not a measurement but an implication: *given what was consumed, what do the specifications imply the pool looked like?* That is the question this paper answers, and the distinction between it and a measurement is

maintained throughout, because the value of the exercise depends entirely on not blurring it.

1.2 Why the state is the unit

Blending is not uniform across the United States. Renewable diesel adoption is concentrated where low-carbon fuel policy rewards it; biodiesel blending is concentrated where feedstock and state mandates favour it; much of the country blends little of either. A national average therefore describes no actual fuel pool.

The state is the finest geography at which consumption by component is published, and it is also the geography at which the relevant policies operate. It is a compromise — fuel is not consumed where it is sold, and a state is not a market — but it is the finest compromise the public record supports.

1.3 Propagating specifications rather than midpoints

The methodological core of this paper is the decision to carry whole envelopes rather than midpoints.

A specification is a permitted range, not a value. Treating FAME’s density as 880 kg/m³ because that is the midpoint of 860–900 kg/m³ produces a point estimate with no indication that the permitted range overlaps the certification fuel’s, and therefore no indication that the *sign* of the computed deviation is an artefact of the midpoint choice.

Carrying the envelopes makes the difference visible. For HVO the interval excludes zero and the direction is established. For FAME it does not and the direction is not. Both results are reported, and the second is reported as prominently as the first, because a paper that reported only the robust half would leave a reader believing something the data does not support.

1.4 Contributions

1. A state-year panel of estimated diesel pool composition and properties for 51 jurisdictions over 14 years (728 rows), with every property estimate carrying a sensitivity interval propagated from the full specification envelopes.
2. An explicit **sign-robustness** determination per state-year and per property, separating what the specifications establish from what the midpoints merely suggest.
3. A correction to the reference fuel: divergence is measured against the EPA certification fuel rather than a European automotive standard. Section 3.2 records what changed when this was corrected, because it changed the headline.
4. Documentation and detection of the EIA 2021 distillate accounting break, which the intuitive construction gets wrong in a way that produces a plausible and false trend.
5. A reproducible, openly licensed pipeline in which every number in this manuscript is interpolated from a machine-written statistics file.

1.5 What this paper does not claim

It does not report measurements. Nobody sampled a pump. Any use of these values as assay results is misuse, and the point is made again in Section 6 because it is the failure mode most likely to occur.

It does not claim that a divergent pool produces divergent emissions. That would require in-service emissions data, which do not exist publicly, and an engine model, which this is not.

It does not claim a direction for FAME-blending states. The specifications do not determine one.

2 Data

2.1 Source

The Energy Information Administration’s State Energy Data System publishes state-level consumption in physical units by fuel and sector. Three series matter here: distillate consumed by the transportation sector, biodiesel, and renewable diesel, each identified by an MSN code.

The panel covers 2011–2024, 51 jurisdictions and 728 state-years, in units of thousand barrels. Source files are not redistributed; they are downloaded by the pipeline and identified in the provenance log by URL, retrieval date, byte count and SHA-256 digest.

2.2 The 2021 accounting break

This deserves its own section because it is the single defect most likely to be reproduced by anyone attempting this analysis independently.

For 1960–2020, the transportation distillate series and the combined distillate-plus-renewables series are **identical in every state and year**. From 2021, EIA’s technical notes state that distillate includes biodiesel and renewable diesel refinery and blender net inputs but excludes their product-supplied consumption.

The consequence is that the intuitive construction — summing the distillate series with the two renewable series to obtain a pool total — is correct before 2021 and double-counts after it. Nationally the overcount averages about 20,008 thousand barrels a year, or 1.726% of the pool. It is small enough to be plausible and structured enough to create a step at exactly the year the accounting changed, which a reader would naturally interpret as a change in the fuel supply.

This pipeline uses the combined series as the denominator and never sums the three. The break is **detected from the data** rather than assumed: the pipeline locates the first year in which the two series diverge and aborts if it is not 2021. An integrity check records the result on every build.

2.3 Component property envelopes

Each pool component carries a density and a cetane envelope taken from its governing specification. These are the paper’s central assumptions and are published as data rather than buried in code.

Table 1. Component property envelopes.

Component	Property	Envelope	Midpoint	Specification
Petroleum diesel (certification envelope)	density	838.9–864.6 kg/m ³	851.75	40 CFR 1065.703 Type 2-D ULSD test fuel; ASTM D975
Petroleum diesel (certification envelope)	cetane	40–50 cetane number	45	40 CFR 1065.703 Type 2-D ULSD test fuel; ASTM D975
FAME biodiesel (B100)	density	860–900 kg/m ³	880	ASTM D6751 / EN 14214
FAME biodiesel (B100)	cetane	47–56 cetane number	51.5	ASTM D6751 / EN 14214
HVO renewable diesel	density	765–800 kg/m ³	782.5	EN 15940 (paraffinic diesel)

Component	Property	Envelope	Midpoint	Specification
HVO renewable diesel	cetane	70–80 cetane number	75	EN 15940 (paraffinic diesel)

The petroleum row is the reference, and its provenance matters: the density envelope is derived from the API gravity range that 40 CFR 1065.703 specifies, not from a commercial diesel standard. ASTM D975 sets no density limit at all, so it cannot serve as a reference even though it is the standard most United States diesel is produced to.

3 Method

3.1 Pool composition and blend properties

For each state-year, the pool is decomposed into petroleum distillate, FAME and HVO shares summing to one, and each property is estimated by volume-weighted linear blending:

$$x_{\text{pool}} = s_p x_p + s_f x_f + s_h x_h$$

where s denotes volume share and x the property value of each component. The deviation from the reference is then

$$\Delta x = x_{\text{pool}} - x_p = s_f (x_f - x_p) + s_h (x_h - x_p)$$

The second form is the one that matters, and it is worth dwelling on. The deviation depends on the component *differences* from petroleum, not on the absolute property values. A common-mode error in the petroleum reference — choosing the wrong point inside its envelope — cancels. This is why the sensitivity analysis in Section 3.3 is computed on the differences rather than on the levels, and it is a correction from an earlier version of this pipeline that computed the band on the absolute value and therefore reported an interval far wider than the analysis actually warranted.

Density blends close to linearly by volume, so the density estimate is reasonable within its stated limits. **Cetane does not blend linearly.** The cetane column is a volume-weighted approximation and is reported as indicative only; it is a further reason not to lean on the cetane result, and it is stated here rather than only in the limitations.

3.2 The reference fuel, and what changed when it was corrected

Divergence is measured against the petroleum component’s specification midpoint, 851.75 kg/m³ and cetane 45.0, derived from 40 CFR 1065.703.

An earlier version of this pipeline used EN 590, the European automotive diesel standard, as the certification-fuel proxy. EN 590 sits roughly 19 kg/m³ lighter than the fuel EPA actually certifies United States engines on. Correcting the reference moved the density sign-robustness result from robust in the large majority of state-years to robust in 2.7% of them, because the corrected reference envelope overlaps FAME’s while the European one did not.

We report this because it is the paper’s most consequential judgement call and because the error was of a specific and instructive kind: the wrong reference was not implausible, it was merely not

the one the question was about, and it produced a stronger headline. A result that strengthens when an assumption is chosen carelessly is the result most in need of the correction being recorded in public.

3.3 Sensitivity: propagating envelopes, not midpoints

For each component and property, the difference from the petroleum reference is evaluated over the **full** admissible range of both, giving an interval rather than a point.

Table 2. Component differences from the certification fuel over full specification envelopes.

Property	Component	Difference from certification fuel	Sign
Density	FAME	-4.6 to +61.1	straddles zero
Density	HVO	-99.6 to -38.9	excludes zero
Cetane	FAME	-3.0 to +16.0	straddles zero
Cetane	HVO	+20.0 to +40.0	excludes zero

The consequence is the paper’s central asymmetry. HVO’s density difference is negative across the entire admissible space and its cetane difference positive across it, so for a state blending renewable diesel the direction holds for any admissible property values and only the magnitude is uncertain. FAME’s intervals straddle zero in both properties, so for a state blending only biodiesel the direction is genuinely undetermined by the specifications.

Each state-year’s deviation interval is computed by evaluating the deviation expression at the combination of component values that minimises it and the combination that maximises it. A state-year is flagged **sign-robust** when that interval excludes zero. The median band width is 2.02 kg/m³ for density and 0.58 cetane numbers; the bands are narrow in absolute terms precisely because most states blend little, and narrow bands around small deviations are still bands that contain zero.

3.4 The divergence index

A single ordering statistic per state-year is reported, combining the normalised density and cetane deviations, so that states can be ranked without choosing a property. It is a convenience for presentation and carries no physical meaning of its own; every claim in this paper is made on the underlying property deviations and their intervals, not on the index.

3.5 Integrity checks

Nine checks run on every build. All pass.

Table 3. Integrity checks.

Check	Result
no_negative_petroleum_share	pass
shares_sum_to_one	pass
pool_total_positive	pass
accounting_break_is_2021	pass
sign_robustness_computed	pass

Check	Result
<code>robustness_reported_in_stats</code>	pass
<code>component_deltas_reported</code>	pass
<code>every_row_has_sensitivity_band</code>	pass
<code>deviation_within_band</code>	pass

These test internal consistency — that shares sum to one, that no share is negative, that the accounting break is where it is expected, that every row carries a sensitivity band and that every reported deviation lies inside its own band. They do **not** test agreement with EIA, which is what the verification gate is for.

Note what the checks deliberately do not include. An earlier version asserted that density direction was robust in more than ninety per cent of state-years, which made the pipeline’s integrity check a test of the conclusion rather than of the data. It has been replaced by checks that the robustness determination was computed and reported, which is a property of the pipeline rather than of the answer.

4 Results

4.1 The national picture

Table 4. Estimated non-petroleum share of the United States transportation distillate pool.

Year	Non-petroleum share
2011	2.16%
2012	2.32%
2013	3.96%
2014	3.76%
2015	4.01%
2016	5.45%
2017	5.25%
2018	4.71%
2019	5.18%
2020	5.83%
2021	5.91%
2022	6.71%
2023	9.56%
2024	11.28%

The share rises over the panel, from 2.16% in 2011 to 11.28% in 2024. The national figure is reported for context only; as Section 1.2 argues, it describes no actual fuel pool.

4.2 The state distribution in 2024

In the latest year, 3 states have a pool estimated lighter than the certification fuel, 40 heavier, and 8 at the reference because they blend essentially nothing. The spread between the lightest and

heaviest state is **38.3 kg/m³**.

That spread is the finding that does not depend on the reference point. Shifting the reference moves every state’s deviation by a constant and leaves the spread unchanged, so a reader who disputes the choice of reference in Section 3.2 can still use the spread.

Table 5. The five state pools with the lowest estimated density deviation, 2024. States blending essentially nothing sit at the reference by construction and appear here with a deviation of zero.

State	Petroleum	FAME	HVO	Blend density	Deviation	Sensitivity interval	Blend cetane	Index
CA	45.4%	3.8%	50.8%	817.6	-34.1	[-50.8, -17.4]	60.5	2.04
OR	63.7%	9.4%	27.0%	835.7	-16.0	[-27.3, -4.8]	53.7	1.07
WA	78.8%	2.7%	18.5%	839.7	-12.1	[-18.6, -5.5]	50.7	0.74
AK	100.0%	0.0%	0.0%	851.8	+0.0	[+0.0, +0.0]	45.0	0.00
MT	100.0%	0.0%	0.0%	851.8	+0.0	[-0.0, +0.0]	45.0	0.00

Table 6. The five state pools with the highest estimated density deviation, 2024.

State	Petroleum	FAME	HVO	Blend density	Deviation	Sensitivity interval	Blend cetane	Index
MN	85.0%	15.0%	0.0%	856.0	+4.2	[-0.7, +9.2]	46.0	0.19
IA	89.6%	10.4%	0.0%	854.7	+3.0	[-0.5, +6.4]	45.7	0.13
IL	92.4%	7.6%	0.0%	853.9	+2.1	[-0.3, +4.7]	45.5	0.10
HI	95.2%	4.8%	0.0%	853.1	+1.4	[-0.2, +2.9]	45.3	0.06
DC	96.5%	3.5%	0.0%	852.8	+1.0	[-0.2, +2.2]	45.2	0.04

4.3 The established result: renewable-diesel states

California is the clearest case and the only one in the panel where the pool is majority non-petroleum. Its 2024 pool is **54.6% non-petroleum**, of which **50.8%** is renewable diesel and **3.8%** biodiesel. The estimated blend density is **817.6 kg/m³**, a deviation of **-34.1 kg/m³** with a sensitivity interval of **[-50.8, -17.4]**. The interval excludes zero, so the direction is established for any admissible property values.

The estimated blend cetane is **60.5**, against a certification fuel envelope of 40–50. The pool’s estimated cetane sits above the entire certification envelope, not merely above its midpoint. Subject to the linear-blending caveat of Section 3.1, this is a state whose fuel is outside the certification specification on two properties at once.

OR, WA follow the same pattern at smaller renewable shares and with correspondingly smaller deviations, and their intervals likewise exclude zero.

4.4 The unestablished result: biodiesel states

The states furthest above the reference in Table 6 are FAME blenders, and every one of them has a sensitivity interval that **contains zero**. The largest, MN, has a midpoint deviation of +4.23 kg/m³ with an interval of [-0.69, +9.16]; the estimate says the pool differs from the certification fuel, and does not say in which direction.

This is not a weakness in the data but a property of the specifications. FAME’s permitted density range (860–900 kg/m³) overlaps the certification fuel’s (838.9–864.6 kg/m³), so a biodiesel batch

at the bottom of its range blended into petroleum at the top of its range moves the pool in the opposite direction from a batch at the top blended into petroleum at the bottom. No amount of consumption data resolves that; only a measurement would.

Across the whole panel the density deviation is sign-robust in **2.7%** of state-years and the cetane deviation in **4.7%**. Those figures are low, and they are the honest figures. They say that this method establishes a direction only where one component’s envelope is disjoint from the reference’s, which is to say only for renewable diesel.

5 Discussion

5.1 What an established direction is worth

For the renewable-diesel states, this paper establishes a directional claim that does not depend on any choice inside the specification envelopes: the pool is lighter and higher-cetane than the fuel the engines burning it were certified on. That is a weaker claim than a measurement and a stronger one than an estimate, and the distinction is worth naming because it is the kind of claim that specification-based analysis is actually good for.

What it supports is a prioritisation argument. If in-service fuel properties are going to be measured — and the companion study’s finding that nobody measures them is the reason they should be — the states where the specifications already imply a divergence are where the first measurements would be most informative.

What it does not support is any statement about emissions. The path from a lighter, higher-cetane fuel to an emissions difference runs through injection, combustion phasing and aftertreatment behaviour, and this paper models none of them.

5.2 Why the unestablished half matters as much

It would have been easy to report only the California result. The FAME finding is negative, awkward, and makes the paper’s headline smaller.

It is also the more transferable result. It says that specification-based inference can establish a direction only when one component’s envelope is disjoint from the reference’s, and that when envelopes overlap, consumption data cannot substitute for measurement no matter how much of it there is. That is a general limit on this method, and any future work using the approach inherits it.

The midpoint estimates for FAME states are not worthless — a midpoint is the expected value under a uniform prior, and refinery practice may well concentrate real fuels near the middle of their ranges. But that is an assumption about the fuel supply, not a consequence of the specifications, and this paper does not make it.

5.3 The spread is the robust national finding

Beyond individual states, the panel’s most durable result is the spread: a range of 38.3 kg/m³ between the lightest and heaviest estimated state pool in 2024. It is robust to the reference choice, because shifting the reference shifts every state by the same constant.

A single national certification fuel is a reasonable construct if the fuel supply is homogeneous. The spread is the quantity that speaks to whether it is, and the national non-petroleum share rose from

2.16% to 11.28% over the panel while remaining concentrated in a small number of states. Whether the spread itself is widening is a question this panel can answer year by year, and we report the 2024 cross-section rather than a trend because the earlier years' component shares rest on smaller volumes and the sensitivity bands scale with them.

5.4 Relationship to the companion study

The companion specification-divergence register asks what the regulation guarantees; this paper asks what the market appears to have supplied. They are complementary and neither substitutes for the other.

The register establishes that density is constrained at certification and not in the market. This paper estimates that the unconstrained property has in fact moved, by different amounts in different states. Together they describe a gap that is both legally open and, on the public evidence, being used — while leaving the question of consequence untouched, because neither has access to a measurement.

6 Limitations

These are the paper's binding constraints, not caveats appended to a finished result.

These are estimates from specification envelopes, not fuel assays. Nobody sampled a pump. Every property here is inferred by combining consumption volumes with published specification ranges. A real tank in a real state contains fuel whose actual properties may sit anywhere inside, or outside, those ranges. The dataset answers “given what was consumed, what do the specifications imply”; it does not answer “what was the density of the fuel at this station on this day”. Any use that treats these values as measurements is misuse.

Component property values are assumptions. Table 1 is the paper. Every deviation scales with those envelopes, and they are the author's first verification item.

Cetane does not blend linearly. The cetane column is indicative only.

Transportation sector only. The analysis uses the transportation consumption series, so off-road, marine, rail, heating and industrial distillate are excluded. Many of the engines in the companion certification panel are nonroad, so the fuel pool those engines actually see is not exactly this one. This is a real mismatch between the two studies and is stated rather than minimised.

State of sale, not state of use. Consumption is attributed to the state of sale; long-haul freight burns fuel across state lines, so a state's estimated pool is not the fuel in the tanks of vehicles operating there.

The reference point is a choice. Section 3.2 records what changed when it was corrected once already. The spread in Section 5.3 is the statistic that survives a different choice.

No in-service emissions claim is supported. The paper stops at fuel properties.

7 Reproducibility and verification status

The pipeline is a sequence of numbered scripts. Provenance is logged at every fetch with a SHA-256 digest. Every number in this manuscript is interpolated from the machine-written statistics file; none is typed by hand.

This package has not passed its verification gate. The verification checklist requires the author to confirm each component property envelope against its governing specification, to reproduce the pipeline from the source files, and to rule on the reference-fuel choice in Section 3.2. Until that attestation exists, the publication gate refuses to remove the draft stamp, and this manuscript must not be cited, deposited or submitted. It is circulated in draft form so that the method can be criticised before the values are relied upon.

Data availability

Processed tables are in the repository under `data/`. The source files are published by the Energy Information Administration and are not redistributed; the pipeline downloads them and records URL, retrieval date, byte count and SHA-256 digest in the provenance log. Component specification envelopes are published as data in `fda_components.csv` with the governing specification cited on each row.

Code availability

The complete pipeline is openly licensed and available in the project repository, including the generator that produced this manuscript.

Competing interests

The author declares no competing interests.

Funding

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Use of AI assistance

Pipeline code, table generation and manuscript drafting were produced with AI assistance. The component property envelopes, the reference-fuel determination and the interpretation of the results are the author's, and the author is responsible for the content.

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Appendix A: component envelopes with source notes

- **Petroleum diesel (certification envelope)**, density (kg/m^3): 838.9–864.6, midpoint 851.75. 40 CFR 1065.703 Type 2-D ULSD test fuel; ASTM D975. Derived from the regulation’s API gravity range 32–37. ASTM D975 sets no density limit.
- **Petroleum diesel (certification envelope)**, cetane (cetane number): 40–50, midpoint 45. 40 CFR 1065.703 Type 2-D ULSD test fuel; ASTM D975. 40 CFR 1065.703 cetane 40–50 by ASTM D613. ASTM D975 requires a minimum of 40.
- **FAME biodiesel (B100)**, density (kg/m^3): 860–900, midpoint 880. ASTM D6751 / EN 14214. EN 14214 range 860–900. ASTM D6751 sets no density limit.
- **FAME biodiesel (B100)**, cetane (cetane number): 47–56, midpoint 51.5. ASTM D6751 / EN 14214. ASTM D6751 minimum 47; EN 14214 minimum 51.
- **HVO renewable diesel**, density (kg/m^3): 765–800, midpoint 782.5. EN 15940 (paraffinic diesel). EN 15940 specified range.
- **HVO renewable diesel**, cetane (cetane number): 70–80, midpoint 75. EN 15940 (paraffinic diesel). EN 15940 minimum 70; typical HVO 70–90.

Appendix B: panel dimensions

- Years: 2011–2024 (14 years)
 - Jurisdictions: 51
 - State-years: 728
 - Volume units: thousand barrels
 - First non-zero year, FAME: 2001
 - First non-zero year, HVO: 2011
 - Accounting break detected at: 2021
 - Identical-regime span: 1960–2020
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Preprint generated 2026-09-15. Every quantity is interpolated from the pipeline’s own statistics file; none is transcribed by hand.